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device lamebus0                # System/161 main bus
device emu* at lamebus*        # Emulator passthrough filesystem
device ltrace* at lamebus*    # trace161 trace control device
device ltimer* at lamebus*    # Timer device
device lrandom* at lamebus*   # Random device
device lhd* at lamebus*       # Disk device
device lser* at lamebus*      # Serial port
#device lscreen* at lamebus*   # Text screen (not supported yet)
#device lnet* at lamebus*     # Network interface (not supported yet)
device beep0 at ltimer*       # Abstract beep handler device
device con0 at lser*          # Abstract console on serial port
#device con0 at lscreen*      # Abstract console on screen (not supported)
device rtclock0 at ltimer*    # Abstract realtime clock
device random0 at lrandom*    # Abstract randomness device

#options net                   # Network stack (not supported)
options semfs                  # Semaphores for userland

options sfs                    # Always use the file system
#options netfs                 # You might write this as a project.

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options dumbvm                # Chewing gum and baling wire.

```

Q1: The VM system configured for Assignment 0 is dumbvm.

Q2: The stack pointer uses Register \$29 in MIPS arch

Q3: OS/161 supports the lamebus bus

Q4: We use `uint32_t` instead of `int` to guarantee a fixed 32-bit size, ensuring portability and correctness in low-level operations like memory addressing, where the size of `int` may vary across architectures.

Q5: Primary handler: `mips_trap()`

Actual termination function: `kill_curthread()`

Q6: Hardclock interrupts are generated at a frequency determined by the `HZ` macro, which is typically set to 100 times per second (100 Hz).

Q7: An SFS (Simple File System) volume name allows up to 32 characters. This is defined by the `SFS_VOLNAME_SIZE` macro in the SFS header files.

Q8: An SFS file has 15 direct blocks. This is defined by the `SFS_NDIRECT` macro.

Q9: The standard abstract file system interface is defined by `struct fs_ops` located in `kern/include/fs.h`. To implement a new file system, we must implement the following four operations:

- `fsop_sync` (Flush dirty buffers to disk)
- `fsop_getvolname` (Return the volume name)
- `fsop_getroot` (Return the root vnode of the filesystem)
- `fsop_unmount` (Unmount the filesystem)

Q10: In modern versions of OS/161 (2.x and later), threads are put to sleep using the `wchan_sleep()` function.

Q11: Process IDs (`pid_t`) in OS/161 are represented as standard 32-bit integers (`int32_t`).

Q12: Operations on a vnode (virtual node) are defined by the `struct vnode_ops` in `kern/include/vnode.h`. There are many operations, but the primary ones include:

- I/O operations: `vop_read`, `vop_write`, `vop_ioctl`

- File metadata: `vop_stat`, `vop_gettype`, `vop_isseekable`, `vop_fsync`
- Directory/Hierarchy operations: `vop_creat`, `vop_lookup`, `vop_remove`, `vop_rename`, `vop_mkdir`, `vop_rmdir`, `vop_link`, `vop_symlink`
- Memory mapping & reclamation: `vop_mmap`, `vop_reclaim`

Q13: The maximum path length is 1024 characters. This is defined by the `PATH_MAX` (or `__PATH_MAX`) macro found in `kern/include/kern/limits.h`.

Q14: The system call number for a reboot is defined by the macro `SYS_reboot`. (`kern/include/kern/syscall.h`) call num is 119

Q15: `STDIN_FILENO` (along with `STDOUT_FILENO` and `STDERR_FILENO`) is defined in the header file at `kern/include/kern/unistd.h`.